OmniWave

School of Data and Information Sciences
The University of North Carolina at Chapel Hill

OmniWave Lab

IEEE International Ultrasonics Symposium 2026

3 Oral Presentations + 2 Posters + Lab Tour

October 4–7, 2026 Raleigh & Chapel Hill, NC

Physics-Guided AI for Computational Wave Imaging

SUNDAY, OCTOBER 4

UNC–Chapel Hill Lab Tour

Campus lab visits: 9:35–11:15 AM and 3:05–4:45 PM

FOUR LIVE DEMOS

AI-Enhanced RF Data Interpolation

Advanced Wave Modeling

Time Reversal and Imaging

Physics-Guided Radiacoustic Imaging

05

MONDAY

ORAL SESSION

Mosha Huang

11:00 AM–12:30 PM

Accurate and Efficient Ultrasonic Wave Simulation Using Split-Exponential Integrator

Room 303 / A2L-05 / Paper 8008

POSTER

Yinan Feng

3:30–4:30 PM

Physics-Constrained Data-Driven Transcranial Ultrasound Full Waveform Inversion with Realistic 3D Acoustic Simulation

Exhibition Hall / A4P-14 / Paper 8417

06

TUESDAY

ORAL SESSION

Yuzhe Guo

8:30–10:00 AM

Improving Speed-of-Sound Reconstruction with Physics-Guided Reverse Time Migration Representations

Room 302 A/B / B1L-03 / Paper 8117

POSTER

Luoyuan Zhang/Titus Spielvogel

10:00–11:00 AM

Assessing Encoder Design for Sparse-to-Dense Ultrasound RF Reconstruction

Exhibition Hall / B2P-15 / Paper 7905

07

WEDNESDAY

ORAL SESSION

Peng Jin

8:30–10:00 AM

End-to-End Transformer for in Vivo Dosimetry Using Quantitative Radiacoustic Imaging

Room 301 A/B / C1L-02 / Paper 8249

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Technical program: Raleigh Convention Center

All times Eastern. Oral listings show full session windows.